Deven Desai

devendesai116@gmail.com | San Jose, CA | +1(669)-292-6556 | LinkedIn | GitHub

SUMMARY

Software Engineering Master's student at San Jose State University and Stanford Hackathon winner specializing in scalable microservice architectures, cloud infrastructure, and low latency backend services. Demonstrated track record at Parspec and Terminal Trends building high throughput applications, optimizing database performance, and deploying containerized services across Python, Node.js, TypeScript, Go, and C++. Proficient in Docker, Kubernetes, AWS, SQL, NoSQL, and RESTful APIs. Eager to leverage microservices expertise and cloud deployment skills to power electric vehicle charging infrastructure at Rivian.

PROJECTS

OmniTrack AI | Cloud Native Real-Time Data Pipeline GitHub

- Built a scalable cloud native backend using Node.js, TypeScript, and Python, orchestrating microservices in Kubernetes for real time data processing.
- Optimized database search performance in PostgreSQL and NoSQL vector stores, achieving sub 100ms query retrieval latencies across large datasets.
- Reduced infrastructure execution overhead by 30 percent by building automated streaming validation and fault tolerant retry mechanisms.

AgenticFlow | Microservice Event Orchestration Platform GitHub

- Designed a high throughput backend microservice architecture using Node.js and Go to manage containerized background tasks deployed on cloud infrastructure.
- Increased backend processing throughput by 40 percent by implementing asynchronous WebSocket interfaces and transactional database models.
- Maintained cross service data consistency by writing comprehensive unit tests and automated CI CD integration workflows.

Low Latency Search Engine | Analytical Indexing Core GitHub

- Queried and indexed millions of structured and unstructured records in real time by engineering an analytical search core in C++ and Java.
- Reduced overall query execution latency by 35 percent through custom memory mapping, low level cache optimization, and efficient data structures.
- Containerized deployment workflows using Docker and deployed cloud services to ensure robust backend scalability.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript, Node.js, Express, Python, Go, C++, Java, FastAPI, RESTful APIs, GraphQL
Cloud & Containerization: Kubernetes, Docker, AWS, Cloud Deployments, Microservices Architecture, CI CD Pipelines
Databases & Analytics: PostgreSQL, MySQL, MongoDB, NoSQL, Redis, Vector Databases (Qdrant), SQL Query Optimization
Software Engineering: Object Oriented Design, System Design, Unit Testing, Code Reviews, Concurrency, Agile Methodologies, Linux

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.